Simulation of CubeSat Detumbling Using B-Dot Controller



Rishav Mani Sharma, Rohit Kawari, Sagar Bhandari, Shishir Panta, Rakesh Chandra Prajapati, and Nanda Bikram Adhikari

Abstract CubeSats tumble with high angular velocity after it is deployed into orbit due to the asymmetric force of deployment. This initial spin is not suitable for precise attitude control. Therefore, it is necessary to reduce the angular velocity of CubeSat to some acceptable value before switching to attitude control strategy. This process of damping spin, known as detumbling is often performed by utilizing the Earth's magnetic field. In this paper, numerical simulation of CubeSat detumbling by using a variant of B-Dot controller with angular velocity feedback is presented. The rotational dynamics of CubeSat is simulated using the RK4 integration of equations of rigid body kinematics and Euler's second law of motion. CubeSat is propagated in low Earth orbit (LEO) using Kepler's equation of orbital motion. The international geomagnetic reference field (IGRF) model is used to find the magnetic field strength in the body frame as the satellite propagates in orbit, spinning on its way. CubeSat is equipped with three magnetorquers, three-axis magnetometer, and three-axis gyroscope.

Keywords CubeSat · Detumbling · B-Dot · Magnetorquer · IGRF model · Simulation

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1 Introduction

The idea of utilizing the Earth's magnetic field to control the attitude of satellites equipped with current-carrying coils was proposed in 1961 [1]. Thirty-eight years later professors Jordi Puig-Suari and Bob Twiggs presented their CubeSat standard and four years after that CubeSats were launched for the first time in 2003 [2, 3]. QUAKESAT 1 (3U) was deployed with passive magnetic control in 2003 and magnetorquers were used for the first time onboard HITSAT (1U) which was launched in 2006 [3]. Most of the CubeSats with mission requirements of active precise attitude control use magnetorquers for detumbling. The main aim of this paper is to present an architecture for simulation of CubeSat detumbling by considering the design parameters of CubeSat as well as magnetorquers.

A magnetorquer is an electromagnet, with or without a ferromagnetic core, that is used as an actuator to detumble the spacecraft using Earth's magnetic field. The magnetic moment generated by magnetorquer interacts with the magnetic field of Earth to produce control torque to the satellite that is used to detumble or control the attitude of CubeSat. This torque relies on the strength of the dipole moment generated by magnetorquer, intensity of Earth's magnetic field at that location in orbit, and relative orientation of these two fields. The generated magnetic torque is maximum if the moment generated by magnetorquer and the field of Earth is perpendicular.

Magnetorquers are suitable actuators due to their simpler construction, higher reliability, lesser weight, and lower cost compared to other class of actuators [4, 5]. Moreover, they do not have moving mechanical parts, do not need propellant, consumes less power and can be operated as long as the solar panel or battery can supply power [5, 6].

This technology has several limitations. Magnetorquers can only generate torque on the plane perpendicular to the magnetic field of the Earth. So they can produce only two control torque components [4]. This shows CubeSat equipped with only magnetorquers is underactuated (components of control torque than the degrees of freedom of satellite) [7]. So, it must be used with other actuators for full three-axis attitude control. Its performance is dependent on the strength of Earth's magnetic field at the orbit, so it is more effective for LEO compared to higher orbits [6]. It has a slow response time compared to actuators such as reaction wheels or propulsion. Hence, it is not a good choice for missions with fast and accurate pointing applications [6].

The actuation of magnetorquer for detumbling requires some control law which must ensure the damping of angular velocity. B-Dot controller is a simple and widely used control algorithm to detumble satellites using magnetorquers. This controller computes the magnitude and direction of current to be passed through magnetorquers in order to detumble the CubeSat. The B-Dot algorithm used in this simulation was proposed by [5] and this particular algorithm was selected due to its proof of global convergence of angular velocities to zero for underactuated spacecrafts. This implies

that the use of this control law detumbles the angular velocity of satellite to zero for any initial values of angular velocity and orientation.

2 Simulation Architecture

The numerical simulation presented in this paper follows the architecture as shown in Fig. 1. The flow of the simulation starts with the generation of the spherical coordinates of CubeSat in ECI frame using the Kepler's equation of orbital dynamics. This coordinate is used by IGRF model to compute the magnetic field of Earth in ECI frame which is then transformed to body frame. Thus, generated magnetic field strength, with the angular velocity obtained using equations of satellite dynamics, is used to implement detumbling control algorithm.

2.1 Mathematical Model of Magnetorquer

The magnetorquer generates a magnetic dipole whose magnitude is proportional to the current passed through it. The magnitude also depends on the number of turns of the coil, its geometry, and the nature of the core used in the coil. The direction of the dipole moment depends on the direction of the current. The mathematical model of both air core and ferromagnetic core magnetorquer is presented.

The mathematical model assumes three identical magnetorquers aligned such that the magnetic moment produced by them align with the principal axes. The magnetic moment, considering air core magnetorquer, is given by the equation below [6],

$$\boldsymbol{\mu}_M = nA\mathbf{i}_M \tag{1}$$

where $\boldsymbol{\mu}_M \in \mathbb{R}^3$ consists magnetic moment generated by each magnetorquer, n is the number of turns of the coil of each magnetorquer, A is the area of the magnetorquer, and $\mathbf{i}_M \in \mathbb{R}^3$ is a vector consisting the magnitude of current through each magnetorquer.

In the case of magnetorquer with ferromagnetic core, Eq. (1) now has an extra term ξ which accounts for the magnetic permeability and the geometry of the core.

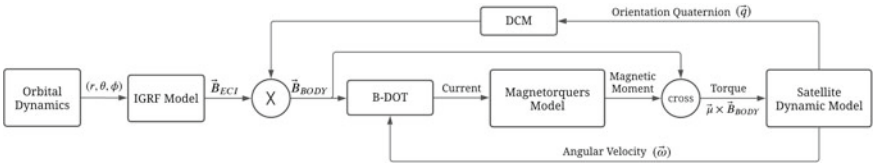


Fig. 1 Architecture of simulation of CubeSat detumbling

$$\boldsymbol{\mu}_M = \xi n A \mathbf{i}_M \tag{2}$$

where ξ is defined as below [8],

$$\xi = \left(1 + \frac{\mu_r - 1}{1 + (\mu_r - 1)N_d}\right). \tag{3}$$

Here, μ_r is the relative permeability of the ferromagnetic core and N_d is demagnetizing factor which is written as [8],

$$N_d = \frac{\left[4 \ln\left(\frac{l_c}{r_c} - 1\right)\right]}{\left(\frac{l_c}{r_c}\right)^2 - 4 \ln\left(\frac{l_c}{r_c}\right)} \tag{4}$$

where l_c is the length and r_c is the radius of core. The torque produced by the magnetorquers due to interaction with Earth's field is given by [9],

$$\boldsymbol{\tau}_M = \boldsymbol{\mu}_M \times \mathbf{B} \tag{5}$$

where $\boldsymbol{\tau}_M \in \mathbb{R}^3$ is the torque due to magnetorquers and $\mathbf{B} \in \mathbb{R}^3$ is magnetic field vector in body frame. Equation (5) models magnetorquers by computing the magnetic torque experienced by the CubeSat using the dipole moment generated by magnetorquers and the local magnetic field of the Earth as input.

2.2 B-Dot Controller

B-Dot controller is a popular control law used to detumble CubeSats. It is written as [9],

$$\boldsymbol{\mu}_{\mathrm{M}} = -k\dot{\mathbf{B}} \tag{6}$$

where $\boldsymbol{\mu}_{\mathrm{M}}$ is a vector with its components corresponding to the magnetic moment command to magnetorquers aligned on each of the principal axes, k is control gain, and $\dot{\mathbf{B}}$ is the derivative of magnetic field vector in body frame. Input current to the magnetorquer is computed using Eq. (1) for air core and (2) for ferromagnetic core magnetorquers, respectively.

This control law offers two major advantages [10]: (1) $\dot{\mathbf{B}}$ is perpendicular to $\mathbf{B}$ for high sensor sampling frequency compared to angular velocity of CubeSat. So the $\boldsymbol{\mu}_{\mathrm{M}}$ produced by magnetorquer will be perpendicular to $\mathbf{B}$, hence, producing maximum control torque. (2) $\dot{\mathbf{B}}$ is proportional to minus the angular velocity vector of satellite $-\boldsymbol{\omega}$. This shows that the control law acts in a way to decrease the angular velocity of the satellite which is in fact the desired effect for detumbling.

The control law used in the simulation is the modified form of B-Dot controller which requires the angular velocity of CubeSat as feedback. Detumbling is straightforward in a fully actuated system as they can perform three-axis control. In such case, the zero angular velocity is asymptotically achieved if the control torque is in the form $\boldsymbol{\tau}_c = -k\boldsymbol{\omega}$ [5]. However, CubeSat with only magnetorquers is underactuated so the same case does not apply to it. It is proved analytically that control law presented in this paper strictly decreases the kinetic energy of CubeSat, which means that it will converge to zero monotonically. This control law is written as,

$$\boldsymbol{\mu}_{\mathrm{M}} = -\frac{k}{||\mathbf{B}||^2}(\mathbf{B} \times \boldsymbol{\omega}) \tag{7}$$

where $\boldsymbol{\omega}$ is the angular velocity vector of the satellite. It can be seen that this equation does not look like the 6 for this control law to be called B-Dot controller in strict sense. However, the authors have also shown this controller being similar to the B-Dot controller where magnetorquer dipole moment command is defined as $\boldsymbol{\mu}_{\mathrm{M}} \propto (\boldsymbol{\omega} \times \mathbf{B}) \approx -\dot{\mathbf{B}}$. Avanzini and Giulietti [5] also presents an equation to find the reasonable value of gain.

$$k = 2n(1 + sin\zeta)I_{\min} \tag{8}$$

where n is mean motion of satellite, ζ is the inclination of the orbit with respect to the geomagnetic equator, and $I_{\min}$ is the value of minimum moment of inertia of the satellite.

2.3 IGRF Model

The IGRF model is used to determine the magnetic field at the coordinate of CubeSat in its orbit. The IGRF is updated every five years and the one used in this simulation is 13th generation (2020–2025). This model computes the strength of magnetic field for given polar coordinates (r, θ, ϕ), where r is the geocentric distance to the satellite, θ is the co-latitude (90-latitude), and ϕ is the east longitude from Greenwich. In simulation, this coordinate is obtained from the Kepler's equation of orbital dynamics. In case of the actual CubeSat on orbit, this information can be obtained from a GPS sensor. The model is implemented based on the detailed algorithm presented in [11] and [12].

The magnetic field strength generated by the IGRF model is in cartesian ECI frame, but the one needed for B-Dot is in body frame. So, coordinate transformation is performed for the magnetic field vector in ECI ($\mathbf{B}_{ECI}$) to body frame ($\mathbf{B}$) using direction cosine matrix constructed using the orientation of satellite in quaternion (q_0, q_1, q_2, q_3)[13].

$$\mathbf{B} = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 + q_0q_2) & 2(q_2q_3 - q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix} \mathbf{B}_{ECI} \quad (9)$$

2.4 Orbit Generation: Keplerian Dynamics

The Kepler's equation is solved to generate the coordinates (r, θ, ϕ) required by IGRF model. The orbit and the position of satellite in the orbit are parameterized using six Kepler elements. They are (1) specific angular momentum (h), (2) inclination (i), (3) right ascension of ascending node (Ω), (4) eccentricity e, and (5) argument of perigee (ω) and true anomaly (θ). It should be noted that the specific angular momentum and true anomaly are often replaced with semimajor axis (a) and mean anomaly (M), respectively [11].

Kepler's equation is used to update the true anomaly which represents the position of satellite as it propagates in its orbit. The equation is,

$$M = E - esinE \quad (10)$$

where M is mean anomaly and E is eccentric anomaly. This transcendental equation is solved for E using Newton's iterative method. Using this value of E, true anomaly (θ) is computed using the relation,

$$tan^2\frac{E}{2} = \sqrt{\frac{1-e}{1+e}}tan\frac{\theta}{2} \quad (11)$$

It is important to note that the initial value of E is not determined using 10 but 11.

Equations 10 and 11 are run in each time step of simulation, and the evolving values of θ represent the position of CubeSat in orbit. However, the Earth is not perfectly spherical in shape and is bulged in the equator. Hence, the gravity of Earth is not perfectly directed to its center. So, for LEO satellites, the gravitational field is not only dependent on its distance from center of the Earth. The second zonal harmonic is the parameter that quantifies this effect and has dimensionless value of $J_2 = 1.08263 \times 10^{-3}$. This effect causes the right ascension Ω and the argument of periapsis ω to change with time which must be considered for LEO satellites. This change is expressed in equations below,

$$\dot{\Omega} = -\left[\frac{3\sqrt{\mu}J_2R^2}{2(1-e^2)^2a^{\frac{7}{2}}}\right] \quad (12)$$

$$\dot{\omega} = -\left[\frac{3\sqrt{\mu}J_2R^2}{2(1-e^2)^2a^{\frac{7}{2}}}\right]\left(\frac{5}{2}sin^2i - 2\right) \quad (13)$$

where R and μ are the radius and gravitational parameter of the Earth, respectively. a and e are the semimajor axis and eccentricity of the orbit, respectively, and i is the orbit's inclination. The orbit is generated by propagating equations (10), (12), and (13) in each time step of the simulation.

Equations (10), (12), and (13) are used to update the Kepler elements as the satellite moves in orbit. The coordinates that are required is not in terms of Kepler elements but in spherical coordinate in ECI. So, in each time step, the updated Kepler elements are converted to the cartesian position vector of CubeSat in ECI frame. Then, this position vector is transformed into spherical coordinates (latitude, longitude, and height) which is used by the IGRF model as input. Curtis [11] presents a detailed algorithm on these transformations.

Transformation of Kepler elements to the position vector in ECI frame. First of all, the Kepler elements are converted into the position vector in perifocal frame [11].

$$\mathbf{r}_{\mathrm{PQW}} = \frac{h^2}{\mu}\left(\frac{1}{1+ecos\theta}\right)\begin{pmatrix} cos\theta \\ sin\theta \\ 0 \end{pmatrix} \tag{14}$$

Then, the coordinate transformation of position vector of satellite in perifocal frame $\mathbf{r}_{PQW}$ to ECI frame $\mathbf{r}_{ECI}$ is performed using direction cosine matrix Q.

$$\mathbf{r}_{ECI} = Q^T \mathbf{r}_{PQW} \tag{15}$$

The 3-1-3 rotation sequences from ECI to perifocus frame are: right ascension of ascending node (Ω), inclination (i), and argument of perigee (ω). The direction cosine matrix (Q), which maps vectors in ECI into vectors in perifocal frame, is defined as [13] [11],

$$Q = \begin{pmatrix} c\theta_3 c\theta_1 - s\theta_3 c\theta_2 s\theta_1 & c\theta_3 s\theta_1 + s\theta_3 c\theta_2 c\theta_1 & s\theta_3 s\theta_2 \\ -s\theta_3 c\theta_1 - c\theta_3 c\theta_2 s\theta_1 & -s\theta_3 s\theta_1 + c\theta_3 c\theta_2 c\theta_1 & c\theta_3 s\theta_2 \\ s\theta_2 s\theta_1 & -s\theta_2 c\theta_1 & c\theta_2 \end{pmatrix} \tag{16}$$

where $c\theta_i = cos\theta_i$ and $s\theta_i = sin\theta_i$ are the shorthand notations, and $\theta_1 = \Omega$, $\theta_2 = i$, and $\theta_3 = \omega$ are the rotation sequences.

Conversion of position vector in cartesian to spherical coordinates. The position vector in ECI in cartesian coordinates is converted to spherical coordinates (r, θ, ϕ) using the following relations [11].

$$r = |\mathbf{r}_{ECI}| \tag{17}$$

$$\theta = sin^{-1} n \tag{18}$$

$$\phi = \begin{cases} cos^{-1}\left(l/cos\theta\right), & \text{if } m > 0 \\ 2\pi - cos^{-1}\left(l/cos\theta\right), & \text{if } m \leq 0 \end{cases} \tag{19}$$

Here, l, m, and n are the direction cosines of the position vector in ECI frame. r, θ, and ϕ are the geocentric height, latitude, and longitude of the satellite which are input to the IGRF model.

2.5 Satellite Rotational Mechanics

The rotational dynamics of CubeSat is modeled using the rigid body kinematics and the Euler's equation of motion. Both of these are coupled differential equations and are integrated numerically using the RK4 method to find the orientation and the angular velocity of the satellite in the next time step.

Kinematics. The rotation of the CubeSat is numerically simulated by integrating the kinematic differential equation for the quaternions as presented in [13].

$$\begin{pmatrix} \dot{q}_0 \\ \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & -\omega_1 & -\omega_2 & -\omega_3 \\ \omega_1 & 0 & \omega_3 & -\omega_2 \\ \omega_2 & -\omega_3 & 0 & \omega_1 \\ \omega_3 & \omega_2 & -\omega_1 & 0 \end{pmatrix} \begin{pmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{pmatrix} \tag{20}$$

Here, (q_0, q_1, q_2, q_3) is the orientation of the satellite in quaternion, and w_1, w_2, and w_3 are the angular velocities of the CubeSat about each of body axes. This coupled differential equation describes the evolution of the orientation of CubeSat.

Kinetics. The angular velocity of the satellite is propagated using the differential equation obtained from the Euler's second law of motion on the rigid body [13, 14],

$$\dot{\mathbf{H}} = \boldsymbol{\tau}_{ext} \tag{21}$$

Here, $\mathbf{H}$ is the angular momentum in inertial frame and $\boldsymbol{\tau}_{\mathbf{ext}}$ is the external torque acting on the satellite. In this case, the source of torque is magnetorquer, so $\boldsymbol{\tau}_{ext} = \boldsymbol{\tau}_{mt}$, where τ_{mt} is torque generated using magnetorquers. By applying the called transport theorem [13] and simplification, it is obtained as,

$$\dot{\boldsymbol{\omega}} = I^{-1}(-\boldsymbol{\omega} \times I\boldsymbol{\omega} + \boldsymbol{\tau}_{mt}) \tag{22}$$

where I is the inertia matrix of the CubeSat and it is assumed to be diagonal, $\boldsymbol{\omega}$ is the angular velocity about the body axis of the satellite.

Table 1 Orbital parameters, simulation parameters, satellite parameters, and the initial conditions for simulation

Orbital parameters, CubeSat configs and initial conditions	Values	Units
Principal moment of inertia	(0.33, 0.37, 0.35)	kg m^2
Max magnetic moment	2.0	A m^2
No. of turns of coil	2000	Turns
Cross section of coil	0.02	m^2
Relative magnetic permeability of core	5500	–
Length of core	0.0800	m
Radius of core	0.0050	m
Simulation time	0.2684	days
Step time	1	secs
Semimajor axis	6978	km
Eccentricity	0	–
Inclination	56	degrees
Argument of periapsis	0	degrees
Longitude of the ascending node	0	degrees
True anomaly	0	degrees
Initial roll, pitch, and yaw	(0.0, 0.0, 0.0)	rad
Initial angular velocities	(0.6, −0.5, −0.4)	rad/sec

3 Simulation and Results

The simulation is performed with the architecture presented in Fig. 1. The CubeSat is assumed to be at the circular orbit of 56 degrees inclination at the height of 6978 km. The control law (7) is used and the gain is estimated using (8). All of the initial conditions and the parameters are shown in Table 1.

Figure 2 shows the nadir projection of CubeSat on the Earth as it revolves around the Earth.

Figure 3 is the magnetic field as observed from the ECI frame. However, the magnetic field in the body frame as seen in Fig. 4 is more chaotic than the field in ECI. This is because the observer (magnetometer fixed to the satellite body) itself is rotating.

It is seen that the field in the body frame gets less random and periodic as time passes. This is because the angular velocity of the satellite is slowly decreasing with time as shown in Fig. 5.

Figure 5 illustrates the detumbling of CubeSat. The magnitude of the angular velocities in each axis is converging to zero with time. Moreover, it is also seen that the angular velocities are converging to zero as claimed by [5]. This shows that the CubeSat is no longer spinning and is ready for fine pointing control strategy. It is observed that the magnetic field in body frame after the CubeSat detumbles is smooth

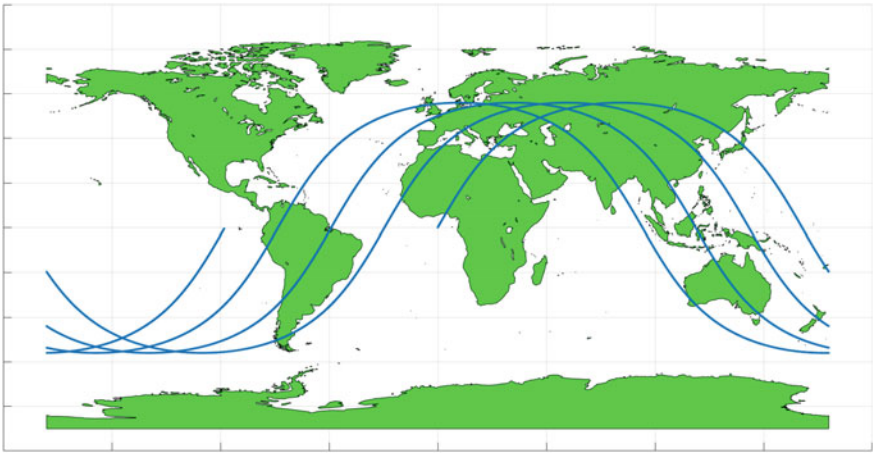


Fig. 2 Groundtrack of the CubeSat

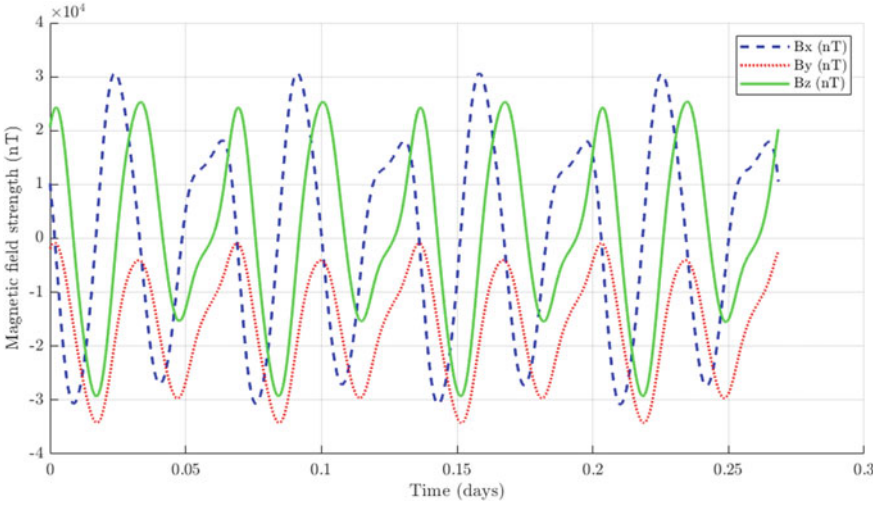


Fig. 3 Magnetic field strength in ECI frame

and periodic. This is because the sensors will measure similar value as the CubeSat travels around the orbit and comes back to same spot without spinning.

Similarly, Fig. 6 shows that the orientation of the CubeSat is gradually being settled to some value as the satellite detumbles. It must be noted that this orientation to is not necessarily the desired orientation. In satellite missions, after successful detumbling, the fine attitude control system is activated to point satellite to the desired orientation.

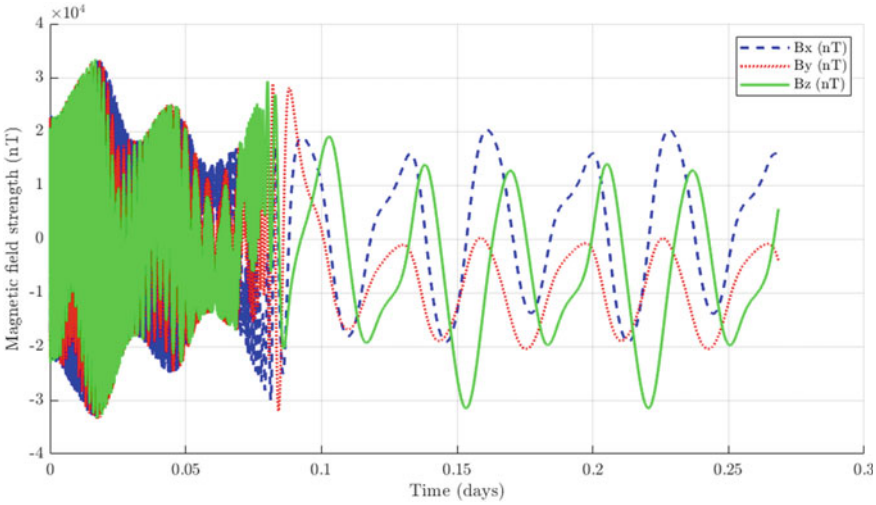


Fig. 4 Magnetic field strength in body frame

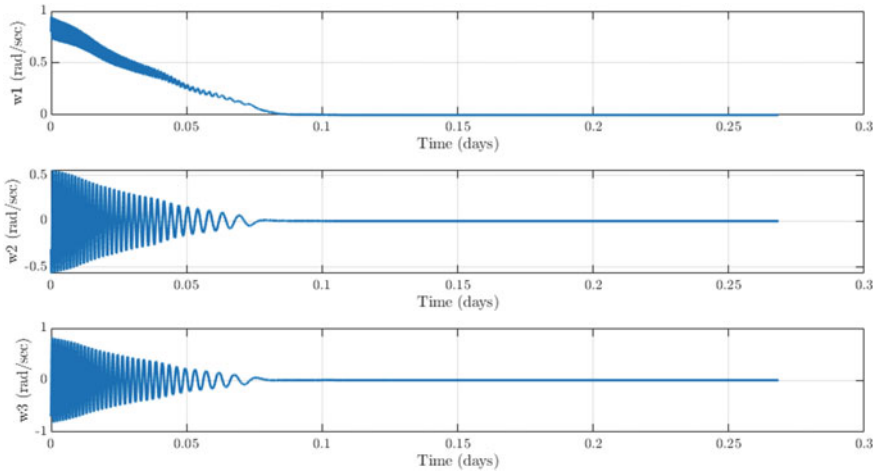


Fig. 5 Angular velocity profile of CubeSat as it detumbles

4 Conclusions

The numerical simulation of magnetic detumbling is performed in a CubeSat spinning on its way around the Earth in circular orbit using a form of B-Dot controller. The CubeSat is fitted with three magnetorquers with ferromagnetic core and three-axis sensors to measure magnetic field and angular velocity. The reason behind selecting this variant of control law is its analytical proof of the global convergence of angular velocity for underactuated satellite. Figure 5 shows the angular velocity of satellite

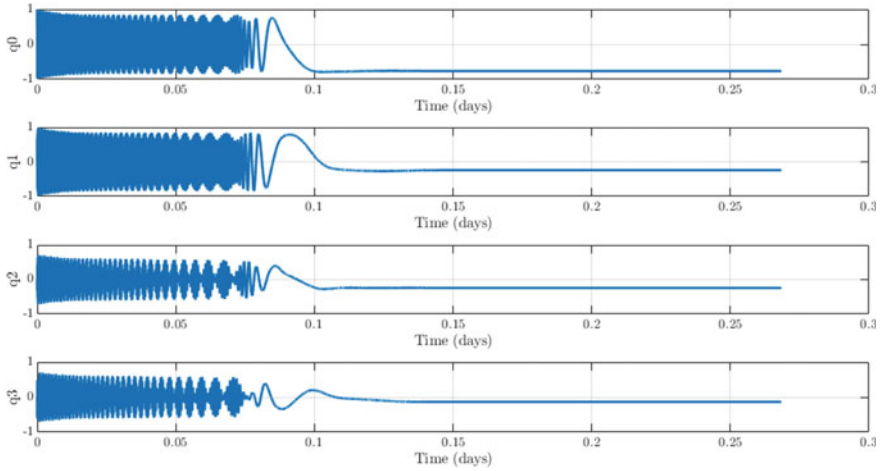


Fig. 6 Orientation profile of the CubeSat in quaternions as it detumbles

converged to zero. The reasonable gain was estimated using (8). To make sure that the implementation of each subsystem is correct, results were compared with published results and online sources. The implementation of orbit generation was verified by generating the ground track of Fig. 4.23 of [11] with given initial conditions in the book. Similarly, the result of IGRF model was compared and verified with the result of online IGRF calculator [15]. The result of transformation of magnetic field strength from ECI to body frame was validated with the result presented in [9].

The further advancement of this research would be hardware implementation of the system on onboard computer of a CubeSat and magnetorquer design using optimization techniques.

References

1. White, J.S., Shigemento, F.H., Bourquin, K.: Satellite Attitude Control Utilizing the Earth's Megnetic Field. Ames Reasearch Center, NASA, Washington (1961)
2. Swartwout, M.: The first one hundred Cubesats, a statistical look. J. Small Satellites **2**(2), 213–233 (2013)
3. Polat, H.C., Virgili-Llop, J., Romano, M.: Towards the thousandth Cubesat, a statistical overview. J. Small Satellites **5**(3), 513–530 (2019)
4. Celani, F.: Robust three-axis attitude stabilization for inertial pointing spacecraft using magnetorquers. Acta Astronaut. **107**(3), 87–96 (2015)
5. Avanzini, G., Giulietti, F.: Magnetic detumbling of a rigid spacecraft. J. Guid. Control Dyn. **35**(4), 1326–1334 (2012). https://doi.org/10.2514/1.53074
6. Bellini, N.: Magnetic actuators for nanosatellite attitude control. Master's thesis (2014)

7. Coverstone-Carroll, V.: Detumbling and reorienting underactuated rigid spacecraft. J. Guid. Control Dyn. **19**(3), 708–710 (1996). https://doi.org/10.2514/3.21680
8. Mehrjardi, M.F., Mirshams, M.: Design and manufacturing of a research magnetic torquer rod. Contemp. Eng. Sci. **3**, 227–236 (2013)
9. Monkell, M., Montalvo, C., Spencer, E.: Using only two magnetorquers to de-tumble a 2u Cubesat **62**, 3086–3094 (2020). https://doi.org/10.1016/j.asr.2018.08.041
10. Leomanni, M.: Comparison of control laws for magnetic detumbling. Research Gate (2012)
11. Curtis, H.D.: Orbital Mechanics for Engineering Students. Butterworth-Heinemann, Oxford (2020)
12. Davis, J.: Mathematical modeling of earth's magnetic field (2010). https://hanspeterschaub.info/Papers/UnderGradStudents/MagneticField.pdf
13. Schaub, H., Junkins, J.L.: Analytical Mechanics of Aerospace Systems. American Institute of Aeronautics and Astronautics Inc, Virginia, USA (2009)
14. Wise, E.D.: Design, analysis, and testing of a precision guidance, navigation, and control system for a dual-spinning Cubesat. Master's thesis (2013)
15. International geomagnetic reference field (IGRF), 13th generation calculator. http://www.geomag.bgs.ac.uk/data_service/models_compass/igrf_calc.html